

yolo_ros

提供了一个基于PyTorch-YOLO的[PyTorch-YOLO](#)的ROS功能包

运行环境：

- 20.04
- ROS Noetic
- Python>=3.7.0环境, PyTorch>=1.7

环境配置：

1. 先安装符合对应的python和pytorch版本的环境

2. 然后安装以下依赖。

```
pip install ultralytics  
pip install rospkg
```

3. 安装Yolo_ROS

```
cd catkin_ws/src  
git clone https://github.com/HT-hlf/yolo_ros.git  
cd ..  
catkin_make
```

使用

使用YOLO检测图像中物体

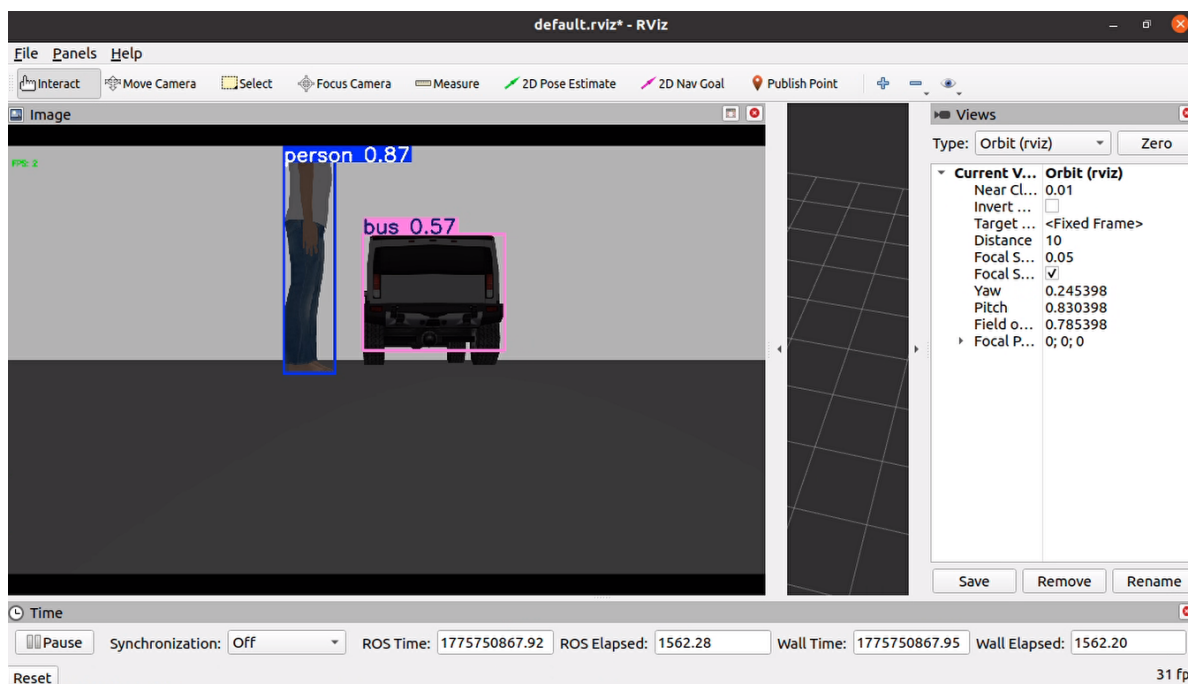
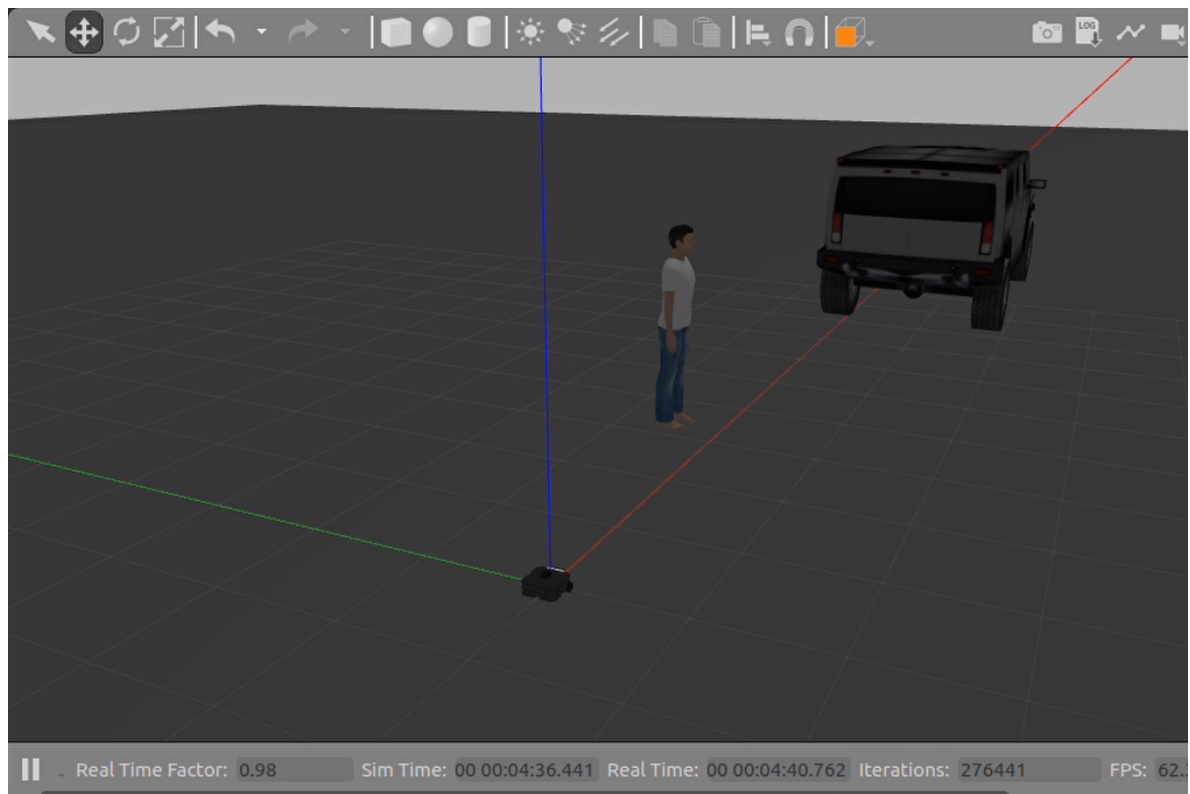
仿真环境中使用

- 启动仿真环境和机器人

```
source devel/setup.bash  
roslaunch turtlebot3_gazebo turtlebot3_empty_world.launch
```

- 启动ros_yolo节点

```
source devel/setup.bash  
roslaunch yolo_ros yolo.launch
```



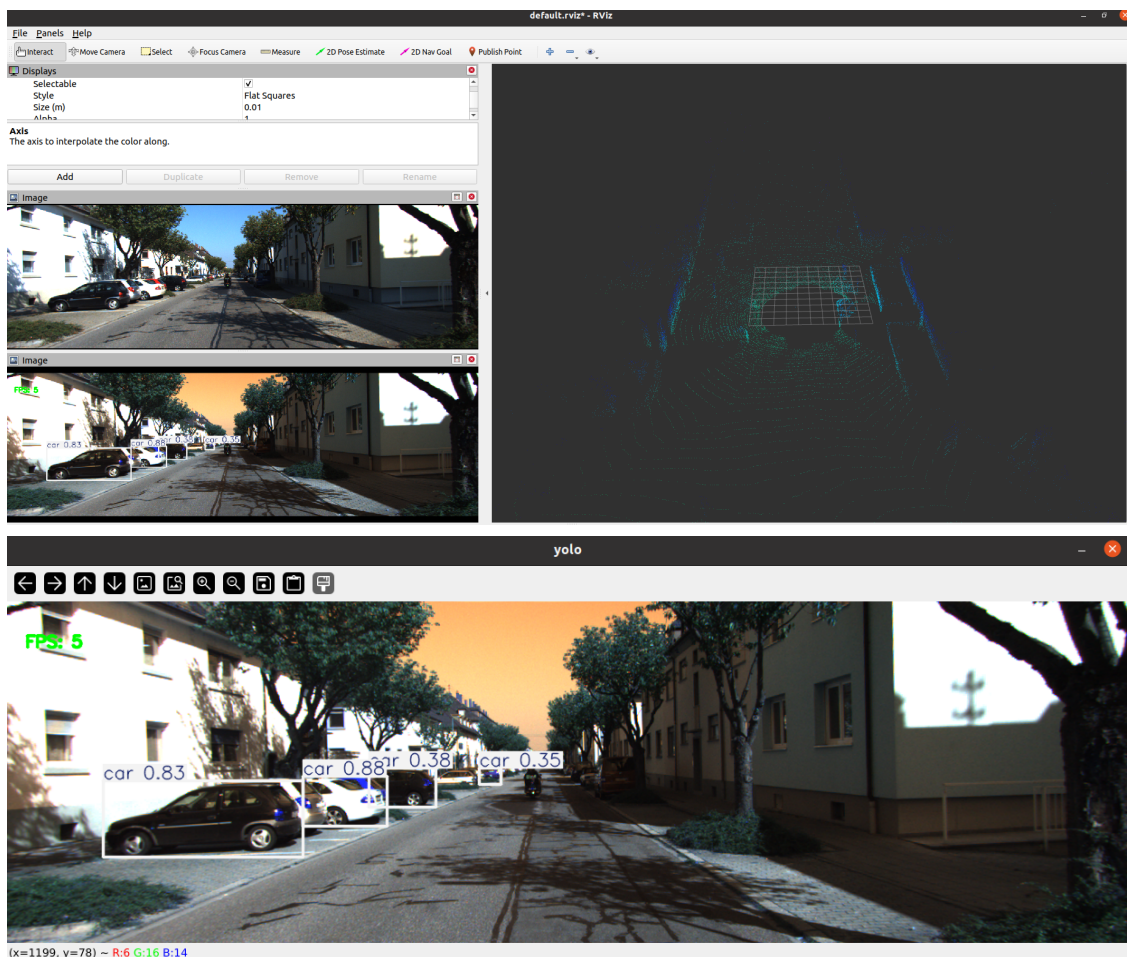
录制数据集中使用

- 播放bag数据

```
rosbag play kitti.bag -r 0.2
```

- 启动ros_yolo节点

```
source devel/setup.bash
roslaunch yolo_ros yolo_kitti.launch
```



使用YOLO检测图像中物体并计算其三维坐标

- 启动仿真环境和机器人

```
source devel/setup.bash
roslaunch turtlebot3_gazebo turtlebot3_empty_world.launch
```

- 启动ros_yolo3D节点

```
source devel/setup.bash
roslaunch yolo_ros yolo_3D.launch
```

